1.

(1) 令 $f(t)=\mathbb{E}[\mathrm{e}^{tX_1}]/\mathrm{e}^{tq}$, 则

$$\begin{aligned}\ln f(t)&=\ln\frac{(1-p)+p\mathrm{e}^t}{\mathrm{e}^{tq}}\\&=\ln((1-p)+p\mathrm{e}^t)-tq.\end{aligned}$$

令 $f'(t)=\frac{p\mathrm{e}^t}{(1-p)+p\mathrm{e}^t}-q=0$, 有 $p\mathrm{e}^t=q(1-p)+qp\mathrm{e}^t$, 即 $t^*=\ln\frac{q(1-p)}{p(1-q)}$, 易知这是 f 的最小值点, 代回就有

$$\Pr\left[\frac{X_1+\cdots+X_n}{n}\geq q\right]\leq\exp(n\ln f(t^*))=\exp(-nd(q\parallel p)).$$

(2) 只需证明 $\mathbb{E}_{X\sim P_i}[\mathrm{e}^X]\leq\mathbb{E}_{X\sim\mathrm{Bern}(p)}[\mathrm{e}^X]$. 这是因为

$$\mathbb{E}_{X\sim P_i}[\mathrm{e}^X]\leq\mathbb{E}_{X\sim P_i}[1+x(\mathrm{e}-1)]=1+p(\mathrm{e}-1)=\mathbb{E}_{X\sim\mathrm{Bern}(p)}[\mathrm{e}^X].$$

(3) 设 $Y=X_2+\cdots+X_n$, 只需要说明 $\mathbb{E}[\mathrm{e}^{X+Y}]\leq\mathbb{E}[\mathrm{e}^X]\mathbb{E}[\mathrm{e}^Y]$, 结合 (1)(2) 的结论就能归纳出目标. 而

$$\begin{aligned}\Delta&:=\mathbb{E}_{X\sim P_X}[\mathrm{e}^X]\mathbb{E}_{Y\sim P_{Y|X}}[\mathrm{e}^Y]-\mathbb{E}_{X,Y\sim P_{XY}}[\mathrm{e}^{X+Y}]\\&=(1-p+p\mathrm{e})\left((1-p)\mathbb{E}_{Y\sim P_{Y|X=0}}[\mathrm{e}^Y]+p\mathbb{E}_{Y\sim P_{Y|X=1}}[\mathrm{e}^Y]\right)\\&\quad-\left((1-p)\mathbb{E}_{Y\sim P_{Y|X=0}}[\mathrm{e}^Y]+p\mathbb{E}_{Y\sim P_{Y|X=1}}[\mathrm{e}^{Y+1}]\right)\\&=(1-p)(-p+p\mathrm{e})\mathbb{E}_{Y\sim P_{Y|X=0}}[\mathrm{e}^Y]+p(1-p)(1-\mathrm{e})\mathbb{E}_{Y\sim P_{Y|X=1}}[\mathrm{e}^Y]\\&=p(1-p)(\mathrm{e}-1)(\mathbb{E}_{Y\sim P_{Y|X=0}}[\mathrm{e}^Y]-\mathbb{E}_{Y\sim P_{Y|X=1}}[\mathrm{e}^Y]).\end{aligned}$$

根据组合意义, 显然 $\mathbb{E}_{Y\sim P_{Y|X=0}}[\mathrm{e}^Y]\geq\mathbb{E}_{Y\sim P_{Y|X=1}}[\mathrm{e}^Y]$, 所以 $\Delta\geq0$, 明所欲证.

2.

在 Sanov bound 中, 先研究 $f(Q):=D(Q\parallel P)$ 在 $\mathbb{E}_{X\sim Q}[X]\geq b$ 的下确界, 以 λ 约束 $1-\sum_i q_i=0$, 以 μ 约束 $b-\mathbb{E}_{X\sim Q}[X]=0$, 则

$$\mathcal{L}(q_1,\cdots,q_n,t,\lambda,\mu)=\sum_{i=1}^n q_i(\ln q_i-\ln p_i-\lambda)+\lambda+\mu\left(b-\sum_{i=1}^n q_iv_i\right).$$

求偏导得

$$\frac{\partial\mathcal{L}}{\partial q_i}=\ln q_i+1-\ln p_i-\lambda-\mu v_i.$$

因此 $\mathcal{L}$ 的驻点满足 $q_i=p_i\mathrm{e}^{\lambda-1}\mathrm{e}^{\mu v_i}$, 这时

$$\sum_{i=1}^n q_i=\mathrm{e}^{\lambda-1}\sum_{i=1}^n p_i\mathrm{e}^{\mu v_i}=\mathrm{e}^{\lambda-1}\mathbb{E}_{X\sim P}[\mathrm{e}^{\mu X}]=1,$$

要求

$$\lambda=1-\ln\mathbb{E}_{X\sim P}[\mathrm{e}^{\mu X}],$$

因此

$$q_i = \frac{p_i \mathrm{e}^{\mu v_i}}{\mathbb{E}_{X\sim P}[\mathrm{e}^{\mu X}]}.$$

由参数 μ 确定, 记为 Q_μ. 代回 KL 散度中:

$$\begin{aligned} D(Q_\mu \parallel P) &= \sum_{i=1}^n q_i \ln \frac{q_i}{p_i} \\ &= \sum_{i=1}^n \frac{p_i \mathrm{e}^{\mu v_i}}{\mathbb{E}_{X\sim P}[\mathrm{e}^{\mu X}]} \ln \frac{\mathrm{e}^{\mu v_i}}{\mathbb{E}_{X\sim P}[\mathrm{e}^{\mu X}]} \\ &= \mu b - \ln \mathbb{E}_{X\sim P}[\mathrm{e}^{\mu X}]. \end{aligned}$$

其中 $b = \mathbb{E}_{X\sim Q_\mu}[X]$ 时 $D(Q_\mu \parallel P)$ 有驻点.

另一面, 在 Chernoff bound 中, 有与之类似的结构 $f(t) := tb - \ln \mathbb{E}_{X\sim P}[\mathrm{e}^{tX}]$, 我们希望在给定 $b > \overline{v}$ 是最大化 f. 求导恰好有

$$f'(t) = b - \frac{\sum_{i=1}^n p_i v_i \mathrm{e}^{tv_i}}{\mathbb{E}_{X\sim P}[\mathrm{e}^{tX}]} = b - \mathbb{E}_{X\sim Q_t}[X], \quad f''(t) < 0.$$

所以 f 上凸, 唯一最大值点处有 $b = \mathbb{E}_{X\sim Q_t}[X]$, 对相同的 b, 只能 $t = \mu$. 这时

$$D(Q^* \parallel P) = D(Q_\mu \parallel P) = f(t) = \max_t (tb - \ln \mathbb{E}_{X\sim P}[\mathrm{e}^{tX}]) =: g(b).$$

那么总是有

$$(n+1)^T \exp(-ng(b)) \geq \exp(-ng(b)).$$

所以 Chernoff bound 不可能弱于 Sanov bound.

3.

(1) 记 $|x| = x \cdot x$, 即 x 中 1 的个数. $f(x) = [|x| \geq \lceil n/2 \rceil]$ 是一个可行的 $\arg\max$, 此时 $\sum_i \mathrm{Influence}_i(f) = \frac{1}{2^{n-1}} \binom{n}{\lceil n/2 \rceil} \lceil n/2 \rceil$. 考虑图 $G = (V, E)$, 其中 $V = \{0,1\}^n$, $(x, y) \in E \Leftrightarrow x \prec y$, 则 $T = \{x : f(x) = 1\}$, $S = V \setminus T$ 是 $0 := (0, \cdots, 0)$ 和 $1 := (1, \cdots, 1)$ 之间的割,

$$\begin{aligned} \sum_i \mathrm{Influence}_i(f) &= \frac{1}{2^n} \sum_x \sum_i (f(x) \neq f(x \oplus e_i)) \\ &= \frac{1}{2^{n-1}} \sum_{(x,y)\in E} [x \in S][y \in T] \\ &= \frac{1}{2^{n-1}} \mathrm{cut}(S, T). \end{aligned}$$

所以我们只需要找到 G 上结点 0 和 1 之间的最大割. 通过调整法容易证明上述 f 给出的割就是一个最大割.

(2) 不妨设 $n = bs$ (忽略常数时容易调整), $f_{b,s}(x) := \bigvee_{i=1}^b \bigwedge_{j=1}^s x[(i-1)s + j]$, 取

$$s = \log_2 n - \log_2 \ln n, \quad b = \frac{n}{s},$$

这样

$$
\begin{aligned}
\frac{1}{2^n}\sum_x f_{b,s}(x) &= 1-\frac{(2^s-1)^b}{2^n}\\
&= 1-\frac{\left(\frac{n}{\ln n}-1\right)^{\frac{n}{\log_2 n-\log_2\ln n}}}{2^n}\\
&= 1-\left(1-\frac{\ln n}{n}\right)^{\frac{n}{\log_2 n-\log_2\ln n}}\\
&= 1-\frac{1}{\mathrm{e}}+o(1).
\end{aligned}
$$

我们相信存在合适的微调使得 $\frac{1}{2^n}\sum_x f_{b,s}(x)\approx\frac{1}{2}$. 这时

$$
\max_i \mathrm{Influence}_i(f_{b,s}) = \frac{(2^s-1)^{b-1}}{2^n} = \frac{1}{2(2^s-1)} = \Theta\left(\frac{\log n}{n}\right).
$$

4.

在 $\mathbb{Z}_2^n$ 上 $\chi:\mathbb{Z}_2^n\to\mathbb{Z}_2^n\to\{\pm1\}\subset\mathbb{R}$, 可以忽略共轭.

(1)

$$
\widehat{g_s}(x) = \frac{1}{2^n}\sum_y f(y\oplus s)\chi_y(x) = \frac{1}{2^n}\sum_y f(y)\chi_y(x)\chi_s(x) = (-1)^{x\cdot s}\hat{f}(x).
$$

(2)

$$
\widehat{g_y}(x) = \frac{1}{2^n}\sum_z (-1)^{z\cdot y} f(z)\chi_z(x) = \frac{1}{2^n}\sum_z f(z)\chi_z(x\oplus y) = \hat{f}(x\oplus y).
$$

(3)

$$
\widehat{f_i}(x) = \frac{1}{2^n}\sum_y (f(y)-f(y\oplus e_i))\chi_y(x) = \hat{f}(x)-\widehat{g_{e_i}}(x) = 2[i\in x]\hat{f}(x).
$$

(4)

$$
\begin{aligned}
\widehat{g_k}(x) &= \frac{1}{2^k}\sum_{y\in\mathbb{Z}_2^k} f(y::\{0\}^{n-k})\chi_y(x)\\
&= \frac{1}{2^k}\sum_{y\in\mathbb{Z}_2^k}\sum_{z\in\mathbb{Z}_2^n}\hat{f}(z)\chi_z(y)\chi_y(x)\\
&= \frac{1}{2^k}\sum_{z\in\mathbb{Z}_2^n}\hat{f}(z)\sum_{y\in\mathbb{Z}_2^k}\chi_{x\oplus z}(y)\\
&= \sum_{z\in\mathbb{Z}_2^{n-k}}\hat{f}(x::z).
\end{aligned}
$$

(5)

$$
\begin{aligned}
\widehat{g_a}(x) &= \frac{1}{2^k}\sum_{z\in\mathbb{Z}_2^k}\frac{1}{2^{n-k}}\sum_{y\in\mathbb{Z}_2^{n-k}} f(z::y)\chi_a(y)\chi_z(x)\\
&= \frac{1}{2^n}\sum_{z::y=u\in\mathbb{Z}_2^n} f(u)\chi_u(x::a)\\
&= \hat{f}(x::a).
\end{aligned}
$$

(6) $\mathrm{Var}[f(X)] = \mathbb{E}[f(X)^2] - \mathbb{E}[f(X)]^2$, 分别计算:

$$\mathbb{E}[f(X)] = \frac{1}{2^n}\sum_x f(x) = \frac{1}{2^n}\sum_x f(x)\chi_0(x) = \hat{f}(0),$$
$$\mathbb{E}[f(X)^2] = \frac{1}{2^n}\|f\|_2^2 \overset{\text{Parseval's}}{=} \|\hat{f}\|_2^2.$$

所以

$$\mathrm{Var}[f(X)] = \|\hat{f}\|_2^2 - \hat{f}(0)^2.$$

5.

$$\begin{aligned}
\widehat{T_\rho(f)}(y) &= \frac{1}{2^n}\sum_x T_\rho(f)(x)\chi_x(y) \\
&= \frac{1}{2^n}\sum_x\sum_z \rho^{|z|}(1-\rho)^{n-|z|} f(x\oplus z)\chi_x(y) \\
&= \frac{1}{2^n}\sum_{k=0}^n \binom{n}{k}\rho^k(1-\rho)^{n-k}\sum_{|z|=k}\sum_x f(x\oplus z)\chi_x(y) \\
&= \frac{1}{2^n}\sum_{k=0}^n \binom{n}{k}\rho^k(1-\rho)^{n-k}\sum_x f(x)\sum_{|z|=k}\chi_{x\oplus z}(y) \\
&= \frac{1}{2^n}\sum_{k=0}^n \binom{n}{k}\rho^k(1-\rho)^{n-k}\sum_x f(x)\chi_x(y)\sum_{|z|=k}\chi_z(y) \\
&= \hat{f}(y)\left(\prod_{i=1}^n((1-\rho)+\rho u_i)\right)\Bigg|_{u_i=(-1)^{[i\in y]}} \\
&= (1-2\rho)^{|y|}\hat{f}(y).
\end{aligned}$$

6.

[UPD 26/4/1] 本题解法有本质错误.

设 $P_C = \mathrm{Unif}(\{0,\cdots,B-1\})$, B 待定, $M = n\cdot 2^B$, $P_S(s) = \Pr\left[\sum_{i=1}^n c_i X_i = s\right]$, 则

$$\begin{aligned}
\Delta(S, S+1) &= \frac{1}{2}\sum_{s=0}^{+\infty}|P_S(s) - P_{S+1}(s)| \\
&= \frac{1}{2}\sum_{s=0}^{+\infty}|P_S(s) - P_S(s-1)| \\
&\le \frac{1}{2}\sqrt{\sum_{s=0}^{M}(P_S(s) - P_S(s-1))^2} \\
&\le \sqrt{\frac{1}{2}\sum_{s=0}^{M}P_S(s)^2}.
\end{aligned}$$

希望

$$\Delta(S, S+1) \le 2^{-\lambda} \Leftarrow \sum_{s=0}^{M}P_S(s)^2 \le 2^{1-2\lambda} \quad \text{w.p. } 1-2^{-\lambda}.$$

计算期望 (以下 $c := (c_1, \cdots, c_n) \sim (P_C)^n, X, X' \sim \mathrm{Bern}(1/2)^n$, $Y \sim \left\{-1 \mapsto \frac{1}{4}, 0 \mapsto \frac{1}{2}, +1 \mapsto \frac{1}{4}\right\}^n$):

$$
\begin{aligned}
\mathbb{E}_c\left[\sum_{s=0}^{M} P_S(s)^2\right] &= \Pr_{c,X,X'}\left[\sum_{i=1}^{n} c_i X_i = \sum_{i=1}^{n} c_i X_i'\right] \\
&= \Pr_{c,Y}\left[\sum_{i=1}^{n} c_i Y_i = 0\right] \\
&= \Pr_{Y}[Y = 0] + \Pr_{c,Y}\left[\sum_{i=1}^{n} c_i Y_i = 0 \mid Y \neq 0\right] \\
&= \frac{1}{2^n} + \mathbb{E}_Y\left[\mathbb{E}_c\left[\sum_{i \neq j} c_i Y_i = -c_j Y_j\right] \mid \exists j\,(Y_j \neq 0)\right] \\
&= \frac{1}{2^n} + \mathbb{E}_Y\left[\mathbb{E}_{\hat{c}_j}\left[\mathbb{E}_{c_j}\left[c_j = -Y_j \sum_{i \neq j} c_i Y_i\right]\right] \mid \exists j\,(Y_j \neq 0)\right] \\
&\leq \frac{1}{2^n} + \mathbb{E}_Y\left[\mathbb{E}_{\hat{c}_j}\left[\frac{1}{B}\right] \mid \exists j\,(Y_j \neq 0)\right] \\
&\leq \frac{1}{2^n} + \frac{1}{B}.
\end{aligned}
$$

于是, Markov bound 给出

$$
\Pr_c\left[\sum_{s=0}^{M} P_S(s)^2 \geq 2^{1-2\lambda}\right] \leq 2^{2\lambda-1}\left(\frac{1}{2^n} + \frac{1}{B}\right).
$$

取 $n = 3\lambda$, $B = 2^{3\lambda}$, 我们就有

$$
\Pr_c\left[\sum_{s=0}^{M} P_S(s)^2 \geq 2^{1-2\lambda}\right] \leq 2^{2\lambda-1}\left(\frac{1}{2^{3\lambda}} + \frac{1}{2^{3\lambda}}\right) \leq 2^{-\lambda},
$$

这时

$$
\Pr_c[\mathrm{Err}(c_1, \cdots, c_n) \leq 2^{-\lambda}] \geq 1 - 2^{-\lambda}.
$$